

Statistical & ML Models

HMMs, SVMs, MFCCs applied.

Time-Difference-of-Arrival (TDoA) for localization.

Transformers & Large Models

Generalization

Noise Mitigation

2004 - 2009

2010 - 2015

2015 - 2025

2025 - ...

Foundations

Cepstral features and timing.

Cross-correlation and signal alignment.

Deep Learning

CNNs, ConvMixers, CNN-RNNs, CoAtNet.

>90% accuracy in clean data, poor in noisy settings.